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REM sleep estimation based on autonomic dynamics using R–R intervals

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Abstract

Objective. We developed an automatic algorithm to determine rapid eye movement (REM) sleep on the basis of the autonomic activities reflected in heart rate variations. **Approach.** The heart rate variability (HRV) parameters were calculated using the R–R intervals from an electrocardiogram (ECG). A major autonomic variation associated with the sleep cycle was extracted from a combination of the obtained parameters. REM sleep was determined with an adaptive threshold applied to the acquired feature. The algorithm was optimized with the data from 26 healthy subjects and obstructive sleep apnea (OSA) patients and was validated with data from a separate group of 25 healthy and OSA subjects. **Main results.** According to an epoch-by-epoch (30 s) analysis, the average of Cohen's kappa and the accuracy were respectively 0.63 and 87% for the training set and 0.61 and 87% for the validation set. In addition, the REM sleep-related information extracted from the results of the proposed method revealed a significant correlation with those from polysomnography (PSG). **Significance.** The current algorithm only using R–R intervals can be applied to mobile and wearable devices that acquire heart-rate-related signals; therefore, it is appropriate for sleep monitoring in the home and ambulatory environments. Further, long-term sleep monitoring could provide useful information to clinicians and patients for the diagnosis and treatments of sleep-related disorders and individual health care.

Keywords: REM sleep, autonomic dynamics, heart rate variability, R–R intervals, automatic sleep staging

(Some figures may appear in colour only in the online journal)

1. Introduction

Rapid eye movement (REM) sleep is one of the sleep stages that is classified by the characteristics of physiological signals based on the Rechtschaffen and Kales (R&K) rules (Rechtschaffen and Kales 1968). From the discovery of REM sleep (Aserinski and Kleitmann 1953), its necessity and function have been reported in various studies. REM sleep constitutes approximately 20–25% of sleep and cyclically repeats 4–6 times with an average period of 90–110 min throughout the night (Carskadon and Dement 2011). REM sleep deprivation influences the increase in the amount of REM sleep during recovery nights, which implies that REM sleep is a physiologically essential element for the sleep process (Endo *et al* 1998). As a functional aspect, REM sleep has been hypothesized to be related to procedural memory that assists in the performance of tasks without consciousness (Rasch and Born 2013). There was significant correlation between the increases in the percentage of REM sleep and the efficiency of language learning (De Koninck *et al* 1989, 1990). Some studies have reported a relationship between REM sleep and creativity. REM sleep awakenings resulted in an enhancement in a complex problem-solving task compared with non-REM (NREM) sleep awakenings (Walker *et al* 2002), and the need for creative thinking increased the amount of REM sleep (Lewin and Gombosh 1973). Further, a nap-based experiment revealed that REM sleep is related to emotional memories (Nishida *et al* 2009). REM sleep is also associated with mood and sleep-related disorders. Patients with primary depressive disorder were characterized with an increased percentage of REM sleep and a reduced REM sleep latency, which is the length of time from the beginning of sleep to the first onset of REM sleep (Peterson and Benca 2011). Patients with primary insomnia showed a decreased percentage of REM sleep (Baglioni *et al* 2014), and a short REM sleep latency is also a characteristic of narcolepsy (Guilleminault and Cao 2011). Furthermore, it is known that apnea is more frequent during REM sleep than NREM sleep (Findley *et al* 1985, Okabe *et al* 1994). Conversely, daytime behaviours including drinking, smoking, medicine use, bright exposure on pre-sleep, and so on also influence the appearance and proportion of REM sleep (Redline *et al* 2004, Higuchi *et al* 2005, Carskadon and Dement 2011). Likewise, REM sleep is important not only as a sleep process itself but also as a link with daily life. Therefore, long-term monitoring of REM sleep could provide valuable information for both sleep research and personal health management.

Polysomnography (PSG) is a standard method for examining human sleep. On the basis of physiological signals such as an electroencephalogram (EEG), an electromyogram (EMG), an electrooculogram (EOG), respiratory-related signals, oxygen saturation, and so on, the sleep architecture and sleep-related disorders are identified and diagnosed. According to the American Academy of Sleep Medicine (AASM) guidelines revised in 2012, REM sleep is scored on the basis of an EEG with a low amplitude and mixed frequency, a low chin EMG tone, and REMs recorded with an EOG (Berry *et al* 2012). However, the analysis of the large volume of data recorded throughout the night is a time-consuming and toilsome task for trained experts (Hwang *et al* 2014). In addition, the attachment of numerous sensors could lead to unfamiliar sleep for the patients. Moreover, visiting the hospital for PSG may not only influence the normal sleep process because of the unusual circumstances but also limit

the number of examinations (Chung *et al* 2009). Therefore, PSG is not suitable for long-term sleep monitoring.

Sleep stages are associated with the activities of the autonomic nervous system (ANS) (Verrier and Harper 2011, Bartsch *et al* 2012). Therefore, the characteristics of the related physiological information such as the respiratory rate and heart rate vary according to the sleep stages. During REM sleep, the lung tidal volume decreases, and the respiratory rate exhibits a frequent and irregular pattern compared with that in NREM sleep (Phillipson 1978, Orem 1994, Heinzer and Series 2011). In addition, the heart rate and its variability are also increased because of the fluctuations between the sympathetic and parasympathetic tones during REM sleep (Lanfranchi and Somers 2011). The heart rate variability (HRV) is a measure of the variation in the heartbeat intervals influenced by the sympathetic and parasympathetic activities of the ANS (Malik *et al* 1996). Various HRV parameters calculated with time-, frequency-domain, and nonlinear analyses revealed a significant difference between NREM and REM sleep (Toscani *et al* 1996, Penzel *et al* 2003, Stein and Pu 2012). Moreover, respiratory sinus arrhythmia (RSA), a periodic variation in the heart rate according to the respiratory cycle (Angelone and Coulter 1964), also exhibits different patterns for REM and NREM sleep (Bartsch *et al* 2012). Therefore, REM sleep monitoring is available because the heart rate provides cardio and even respiratory information associated with autonomic nervous activities.

In summary, the long-term monitoring of REM sleep is important because it could offer significant information for healthcare and research. Because the sleep stages are associated with autonomic control, the detection of REM sleep is accessible with ANS-related signals such as the heart rate. In this study, we propose a method for estimating REM sleep using the characteristics of the R–R intervals reflected by the autonomic dynamics. Because the algorithm only requires the heart rate, it is applicable for REM sleep monitoring with wearable and mobile devices that record an electrocardiogram (ECG) or alternative signals such as a photoplethysmogram (PPG), ballistocardiogram (BCG), and so on. Finally, in combination with methods that estimate other stages such as slow-wave sleep (SWS) and wakefulness, the proposed method is expected to monitor the comprehensive sleep architecture in ambulatory and home environments.

2. Methods

2.1. Study subjects

Twenty-one healthy subjects (male: 12, female: 9) and 30 subjects (male: 25, female: 5) with obstructive sleep apnea (OSA) participated in this study. Apnea was scored with an amplitude reduction in the oronasal thermal signal of more than 90% from the pre-event baseline lasting at least 10 s (Berry *et al* 2012). In addition, hypopnea was detected when the event satisfies all of the following criteria: an amplitude drop in the nasal pressure of more than 30% lasting at least 10 s, and a decrease in the oxygen desaturation of more than 3% from the pre-event baseline or the event is related with arousal (Berry *et al* 2012). The apnea hypopnea index (AHI) was calculated by dividing by the number of occurrences of apnea and hypopnea by the hours of sleep (Berry *et al* 2012). The algorithm was developed with the data from 26 subjects with an average AHI of 17.86 ± 19.36 and was validated with the data from 25 separate subjects with an average AHI of 13.78 ± 16.65 . A subject's demographics and sleep-related variables were not significantly different between the training and validation sets, as summarized in table 1. The research was approved by the Institutional Review Board of Seoul National University Hospital (IRB No. H-1405-124-582).

Table 1. Summary of sleep-related variables and demographics.

Variables	Mean $\pm$ S.D.	
	Training	Validation
Gender (male/female)	20/6	17/8
Age (years)	35.21 $\pm$ 15.76	31.71 $\pm$ 17.10
BMI (kg m ⁻²)	23.87 $\pm$ 4.23	23.10 $\pm$ 4.85
Total recording time (min)	495.54 $\pm$ 41.70	497.60 $\pm$ 46.25
Total sleep time (min)	451.90 $\pm$ 51.75	448.72 $\pm$ 62.00
Sleep efficiency (%)	91.15 $\pm$ 6.46	90.35 $\pm$ 10.48
Stage W (%)	8.85 $\pm$ 6.46	9.65 $\pm$ 10.48
Stage N1 and N2 (%)	62.44 $\pm$ 7.53	58.72 $\pm$ 11.60
Stage N3 (%)	9.47 $\pm$ 6.92	12.78 $\pm$ 10.15
Stage REM (%)	20.24 $\pm$ 4.68	18.85 $\pm$ 5.25
AHI (events/hour)	17.86 $\pm$ 19.36	13.78 $\pm$ 16.65

S.D.: standard deviation, BMI: body mass index, W: wakefulness, REM: rapid eye movement, N: NREM, AHI: apnea hypopnea index.

Note: none of the values was significantly different between training and validation sets.

2.2. Data recording

All participants underwent nocturnal PSG at the Center for Sleep and Chronobiology at the Seoul National University Hospital. The data were recorded using NI-DAQ 6221 (National Instruments, Austin, TX, USA) with the following channels: EEGs from the C3-A2 and O2-A1 positions, submental and bilateral tibialis anterior EMGs, bilateral EOGs, chest and abdominal volume changes using piezoelectric sensors, nasal–oral airflow, body position, snoring, an ECG at lead II, and blood oxygen saturation. All data were gathered with sampling rate of 250 Hz. According to an AASM guideline (Berry *et al* 2012), a trained technician scored PSG data from each subject every 30 s (1 epoch). Finally, the scored results were reviewed by sleep clinicians.

2.3. HRV parameter extraction

From the recorded PSG data, an ECG was processed with a high-pass filter at 0.5 Hz and a low-pass filter at 30 Hz (5th order, infinite impulse response Butterworth). Then, the R-peaks were found using a self-developed algorithm (Choi *et al* 2009) and manually corrected. On the basis of the extracted R-peaks, the R–R intervals, the time span between consecutive R-peaks in an ECG, were obtained.

Using the R–R intervals, 14 HRV and respiratory-related parameters were calculated, as summarized in table 2. Many studies have reported that the heart rate increases, is irregular, and fluctuates during REM sleep (Somers *et al* 1993, Trinder *et al* 2001, Lanfranchi and Somers 2011). Thus, the mean of the R–R intervals (mRR), i.e. the inverse form of the mean heart rate, was calculated in each 30 s epoch. Then, the standard deviation of mRR (std_{mRR}) was acquired with a symmetrical sliding window of 5 epochs shifted by an epoch to obtain successive values. r_{mRR} is an HRV parameter representing the time series fluctuation in the R–R intervals. First, the R–R intervals within 30 s were normalized to have a zero mean and unit variance (nRRI). Then, nRRI was smoothed using a Savitzky–Golay finite impulse response (FIR) filter (Savitzky and Golay 1964) with a polynomial order of 2. Finally, the correlation coefficient (r_{nRRI}) was calculated between two sets of R–R intervals. Figure 1(a)

Table 2. Descriptions of the HRV parameters for estimating REM sleep.

Parameter	Window (s)	Description
mRR ^a	30	Mean of RRI
r_{nRR} ^a	30	Correlation coefficient between nRRI and smoothed nRRI
std_f ^a	150	Standard deviation of dfr within adjacent 5 epochs
std_{mRR} ^a	150	Standard deviation of mRR within adjacent 5 epochs
VLF	300	Spectral power of RRI in frequency range between 0.003 and 0.04 Hz
LF	300	Spectral power of RRI in frequency range between 0.04 and 0.15 Hz
HF	300	Spectral power of RRI in frequency range between 0.15 and 0.4 Hz
LFHF	300	LF to HF ratio
nLF	300	LF divided into summation of LF and HF
nHF ^a	300	HF divided into summation of LF and HF
sd_1 ^a	300	Short-term variability of RRI represented by Poincare plot analysis
sd_2	300	Long-term variability of RRI represented by Poincare plot analysis
α_1	300	Short-term scaling exponent of RRI
α_2 ^a	300	Long-term scaling exponent of RRI

RRI: R–R intervals, nRRI: normalized RRI, dfr: dominant frequency of RRI in frequency range between 0.15 and 0.5 Hz.

^a HRV parameters selected by optimizing process for REM detection.

shows an example of nRRI and the smoothed nRRI measured during N3 sleep, and figure 1(b) presents the corresponding data recorded during REM sleep. Figure 1(c) shows the correlation coefficients (r_{nRR}) between nRRI and the smoothed nRRI with the sleep stages. Because the baseline of the R–R intervals highly fluctuates during REM sleep, r_{nRR} increased in REM sleep compared with that in NREM sleep.

REM sleep is also characterized with respiratory dynamics. On the basis of the RSA phenomenon, the respiratory variability was quantified using the R–R intervals. First, the R–R intervals in each epoch were interpolated with the shape-preserving precise cubic method, then high-pass filtered at 0.15 Hz, and sequentially low-pass filtered at 0.5 Hz. The dominant frequency in the range of 0.15–0.5 Hz was extracted from the result of a fast Fourier transform (FFT) with a rectangular window for the processed data. Then, the standard deviation of the dominant frequencies (std_f) was computed with a symmetrical sliding window of 5 epochs by shifting an epoch. In addition, the HRV parameters in the frequency domain were acquired. First, a shape-preserving precise cubic interpolation was applied to the R–R intervals in a time window of 10 epochs (300 s). Then, the spectral powers in the frequency ranges of 0.003–0.04 Hz (VLF), 0.04–0.15 Hz (LF), and 0.15–0.4 Hz (HF) were calculated using an FFT analysis. Consecutive values were computed using the R–R intervals in the corresponding time window shifted by 1 epoch. On the basis of LF and HF, the ratio of LF and HF (LFHF) was calculated, and the normalized LF (nLF) and HF (nHF) were also calculated by dividing by the sum of LF and HF in each epoch.

Nonlinear HRV parameters were also accessed using the R–R intervals. A Poincare plot is a 2D scatterplot between the signal and time-delayed signal, which is used to investigate the

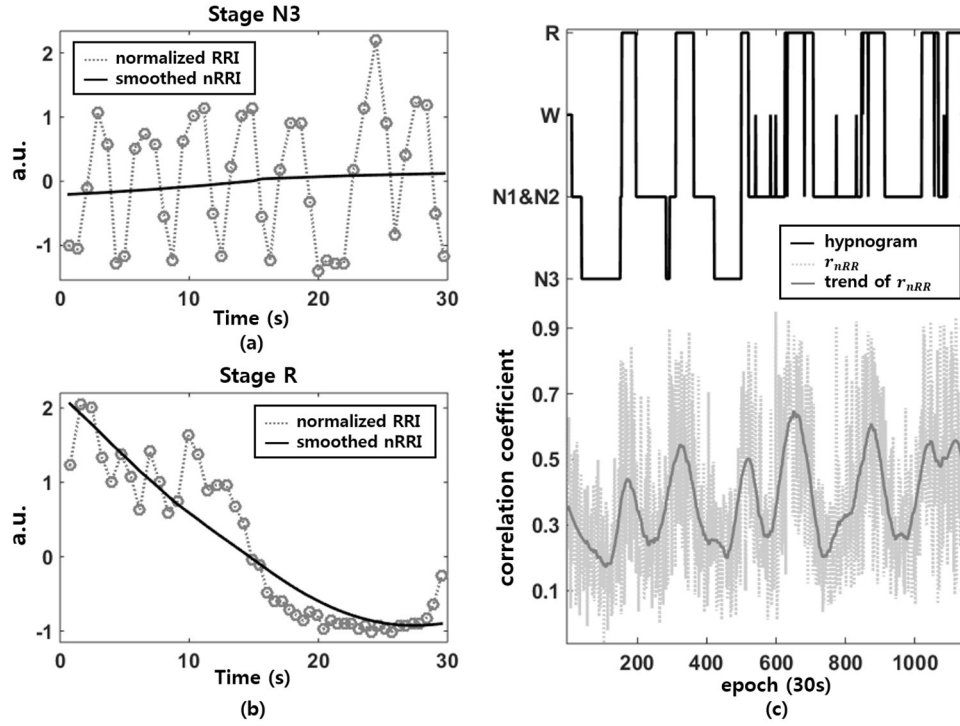


Figure 1. An example of the patterns of R–R intervals and a representative HRV parameter (r_{nRR}) during sleep. (a) Normalized R–R intervals (nRRI) and the smoothed nRRI during N3 sleep, (b) the corresponding data from REM sleep, and (c) the correlation coefficients (r_{nRR}) between nRRI and the smoothed nRRI with the sleep stages.

nonlinear dynamics of time series data (Woo *et al* 1992). It allows quantitative measures of the fluctuations in the heart dynamics with trends (Mourot *et al* 2004). sd_1 and sd_2 are the parameters from the Poincare plot analysis and represent the short- and long-term variabilities in the R–R intervals, respectively (Piskorski and Guzik 2007). Moreover, they provide information for sympathetic and parasympathetic activities (Kamen *et al* 1996, Brennan *et al* 2002). sd_1 and sd_2 were calculated using the following equations:

$$X_1 = \{x_1, x_2, \dots, x_{n-1}\}, \quad X_2 = \{x_2, x_3, \dots, x_n\}, \quad (1)$$

$$Y = \{y_1, y_2, \dots, y_{n-1}\} = \frac{(X_1 - X_2)}{\sqrt{2}}, \quad (2)$$

$$Z = \{z_1, z_2, \dots, z_{n-1}\} = \frac{(X_1 + X_2)}{\sqrt{2}}, \quad (3)$$

$$sd_1 = \sqrt{\frac{1}{n-2} \cdot \sum_{k=1}^{n-1} (y_k - u_Y)^2}, \quad (4)$$

$$sd_2 = \sqrt{\frac{1}{n-2} \cdot \sum_{k=1}^{n-1} (z_k - u_Z)^2}, \quad (5)$$

where x_n denotes the R–R intervals, and n is the number of R–R intervals. Thus, X_1 consists of R–R intervals except for the last R–R interval, and X_2 contains the second-to-last R–R intervals within the time window. Y and Z are the sets equal to the difference between and the sum of X_1 and X_2 divided by the square root of 2, respectively. Finally, sd_1 and sd_2 represent the standard deviations of Y and Z , respectively, where u_Y and u_Z are the means of Y and Z , respectively. Here, the time window was 10 epochs, and the window was shifted by 1 epoch to calculate subsequent values.

A detrended fluctuation analysis (DFA), which is a method for quantifying the long-range correlation in a non-stationary time series (Peng *et al* 1994), has been widely used to assess physiological signals. In several studies, the R–R intervals measured during sleep were analysed using a DFA. The short- and long-term scaling exponents (α_1 and α_2)—quantitative measures of a DFA—showed significant differences among sleep stages, revealing that α_1 and α_2 could provide meaningful information of autonomic nervous activities during sleep (Bunde *et al* 2000, Penzel *et al* 2003). Therefore, α_1 and α_2 were also extracted from the R–R intervals with following equations to establish the model for REM sleep detection:

$$p_j = \sum_{i=1}^j (x_i - \hat{x}), \quad (6)$$

$$F_k = \sqrt{\frac{1}{n} \sum_{j=1}^n (p_j - p_j^k)^2}. \quad (7)$$

First, p_j was computed with the R–R intervals (x_i). j increases from 1 to n , where n is the number of R–R intervals in the time window of 300 s. x_i represents the i th R–R interval, and $\hat{x}$ corresponds to the average of the R–R intervals in the segment. Then, the integrated R–R intervals (p_j) were detrended by the subtraction of the local linear trend calculated using k points of p_j (p_j^k). Here, k increases from 4 to 64. Finally, α_1 and α_2 were expressed as the regression slopes between both the logarithmically scaled F_k and k , where the ranges of k used for the estimation of each slope were 4–16 and 16–64, respectively. Here, the time window was 300 s, and the window was shifted by 30 s to obtain consecutive values. In total, 14 HRV parameters were extracted from the R–R intervals, and an algorithm for detecting REM sleep was developed with these parameters.

2.4. Algorithm description

The algorithm for detecting REM sleep was established using the obtained HRV parameters. Our strategy was (1) to extract the feature that shows a difference between REM and NREM sleep, (2) to generate and apply an appropriate threshold for the feature to determine REM candidates, (3) and to determine REM sleep from the candidates on the basis of the physiological background related to REM sleep. The details of feature extraction, threshold determination, and determination of REM sleep are explained in following sections.

2.4.1. REM feature extraction. Although the HRV during sleep is influenced by various factors such as the environment and sleep-related disorders, a major fluctuation may be caused by the sleep cycle. Here, a principal component analysis (PCA) was applied to extract the feature that varies according to the main sleep cycle. A PCA is a statistical method for transforming a set of measured variables into a set of linearly uncorrelated variables, the so-called principal components (Hotelling 1933). On the basis of a PCA, a set of data is decomposed

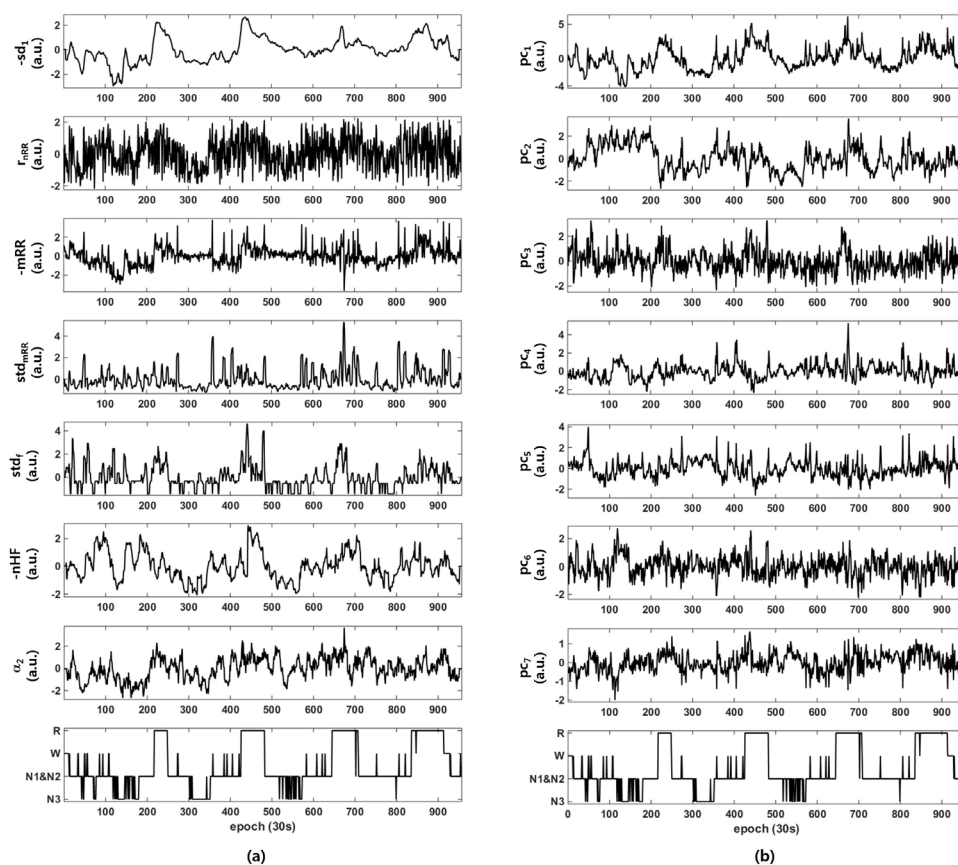


Figure 2. An example of REM feature extraction based on a PCA. (a) An example of 7 HRV parameters (normalized) applied to a PCA and (b) the principal components extracted from the corresponding normalized HRV parameters.

into eigenvectors with eigenvalues and then projected onto each eigenvector scaled by the corresponding eigenvalue. Finally, a set of principal components is sorted in order of variance. Therefore, the first principal component (PC_1) obtained using the HRV parameters represents the major fluctuation reflected by the patterns of a sleep cycle. The HRV parameters were standardized with a zero mean and unit variance and applied to the PCA process. Then, PC_1 was extracted and normalized again to have a zero mean and unit variance (nPC_1) and low-pass filtered with a Savitzky–Golay FIR filter with a polynomial order of 2 (sPC_1). Finally, the result was regarded as the feature for REM sleep detection. However, it is unknown which combination of HRV parameters for the PCA could generate the optimal feature for the determination of REM sleep. Thus, the PCA was recurrently performed for all possible combinations of the HRV parameters using the training data. The optimal combination was selected to produce the feature that maximizes the Cohen’s kappa for REM sleep detection. Figure 2 shows an example of an REM feature extracted from 7 HRV parameters using a PCA. The selected HRV parameters are shown in figure 2(a), and the principal components extracted from the corresponding HRV parameters are presented in figure 2(b). As a result of the PCA, PC_1 reflected the major fluctuation represented in each parameter, and the fluctuation that shows a cyclic pattern corresponded with the REM sleep cycle.

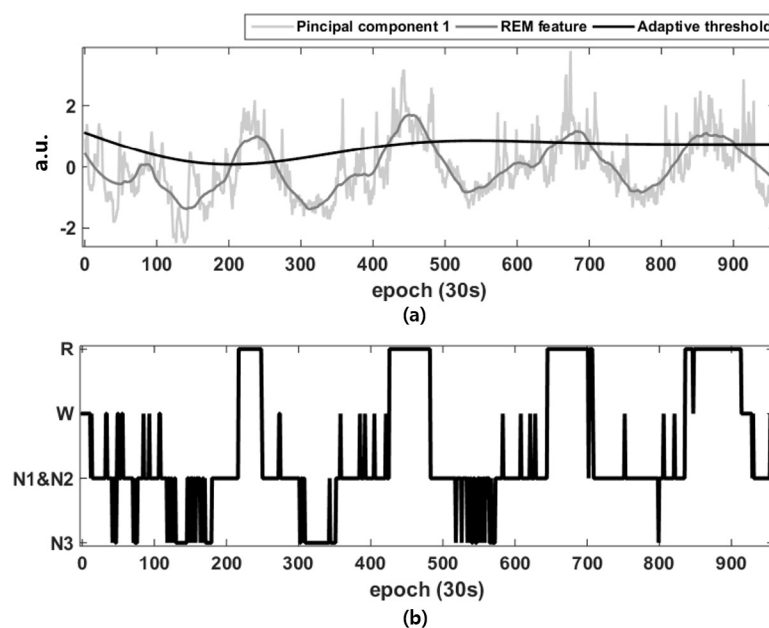


Figure 3. An example of (a) extracted features with the adaptive threshold (principal component 1: PC_1 , REM feature: sPC_1 , and adaptive threshold: $IPC_1 + C_{th}$) and (b) the reference sleep stage.

2.4.2. Adaptive threshold determination. An adaptive threshold for identifying an REM sleep candidate was determined using the acquired REM feature (sPC_1), which was the smoothed first principal component (PC_1) from the HRV parameters. First, a low-pass filter was applied to the extracted feature to find the baseline fluctuation in the REM feature (IPC_1). Here, the sampling frequency was $1/30$ Hz, and the cut-off frequency was determined to be $2/(30 \times m)$ Hz, where m is the total number of sleep epochs, and 30 is time span of an epoch in which the value of an REM feature was calculated. That is, the baseline trend of the REM feature (IPC_1) includes the components that varied for less than 2 cycles during the total recording time. Finally, a constant value (C_{th}) was added to IPC_1 ; then, it was regarded as the adaptive threshold for detecting an REM sleep candidate. That is, the epoch in which sPC_1 is greater than the sum of IPC_1 and C_{th} was designated as an REM sleep candidate. The constant value added to IPC_1 was optimized to maximize Cohen's kappa for REM detection with the data from the training set. Figure 3 shows an example of PC_1 based on the HRV parameters, the final REM feature (sPC_1), and the adaptive threshold (IPC_1 with C_{th}) with the reference sleep stage.

2.4.3. Heuristic rule. During normal human sleep, the average time duration of the first NREM–REM sleep cycle is about 70–100 min (Carskadon and Dement 2011). In detail, sleep starts with the N1 stage and progresses through N2 and N3 sleep before the first REM sleep appears. This implies that REM sleep rarely occurs within approximately an hour after the beginning of sleep. On the basis of this fact, the detected REM sleep candidates within a specific time after the beginning of sleep were considered as NREM sleep. The specific time (C_{hr}) was determined to maximize Cohen's kappa coefficient using the training data.

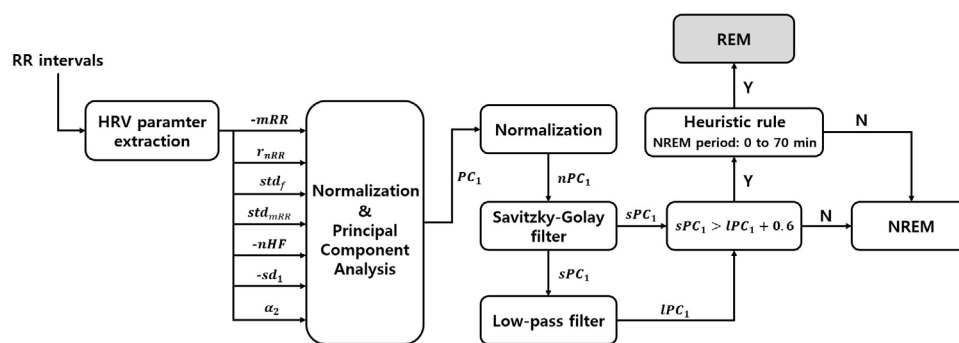


Figure 4. Procedure for REM sleep estimation. The HRV parameters and values for the thresholds and heuristic rule were determined to maximize Cohen's kappa coefficient using training data.

2.5. Performance analysis

The algorithm was developed on the basis of the data from training set of 26 subjects and was evaluated using the data from a separate set of 25 subjects. For reference, two classes were defined as REM and NREM including N1, N2, N3, and wakefulness. To evaluate the performance of REM detection, the accuracy (%), sensitivity (%), and specificity (%) were obtained by comparing the results from the PSG and those from the algorithm according to the epoch-by-epoch analysis. Cohen's kappa coefficient, a measure of inter-rater agreement (Viera and Garrett 2005), was also used to assess the performance of the proposed algorithm. All statistical values were compared between the training and validation sets and between healthy and OSA groups. Moreover, REM sleep-related information such as the percentage of REM sleep (%) and the duration of REM sleep (min) was extracted from the reference PSG and compared with the result of automatic scoring by the proposed method. Moreover, the agreement between the reference of the REM sleep-related information and the result of the algorithm was assessed using a Bland–Altman plot with a significance level of 5%.

3. Results

3.1. HRV parameters and threshold determination

The procedure for REM sleep estimation with the selected HRV parameters, the specific values for the threshold, and the heuristic rule are presented in figure 4. Parameter selection was optimized to maximize Cohen's kappa coefficient with the data from the training set. As presented in table 3, mRR, r_{nRR} , std_f , std_{mRR} , nHF, sd_1 , and α_2 were finally determined as the set of the parameters used for REM feature extraction using a PCA. In addition, the 7 selected parameters were the most frequently used HRV parameters in the optimization process, as shown in figure 5. The adaptive threshold for detecting the candidates of REM sleep was determined with the sum of IPC_1 and a constant value (C_{th}) of 0.6. In addition, the time threshold (C_{hr}) used in the heuristic rule was chosen to be 70 min (140 epochs). That is, the first 70 min after the beginning of sleep were regarded as NREM sleep. The determination of C_{th} and C_{hr} is shown in figure 6.

Table 3. Selected parameters to maximize Cohen's kappa for each combination in the training set.

	Parameters	Cohen's Kappa
1	std_f	0.59 ± 0.13
2	$std_f, -nHF$	0.59 ± 0.10
3	$-mRR, std_f, nLF$	0.62 ± 0.13
4	$-mRR, std_f, -nHF, \alpha_2$	0.62 ± 0.13
5	$-mRR, r_{nRR}, std_f, -nHF, \alpha_2$	0.62 ± 0.13
6	$-mRR, r_{nRR}, std_f, std_{mRR}, -nHF, -sd_1$	0.62 ± 0.12
7	$-mRR, r_{nRR}, std_f, std_{mRR}, -nHF, -sd_1, \alpha_2$	0.63 ± 0.13
8	$-mRR, r_{nRR}, std_f, std_{mRR}, LF, -nHF, -sd_1, \alpha_2$	0.61 ± 0.13
9	$-mRR, r_{nRR}, std_f, std_{mRR}, VLF, nLF, -nHF, -sd_1, \alpha_2$	0.60 ± 0.13
10	$-mRR, r_{nRR}, std_f, std_{mRR}, VLF, LF, nLF, -nHF, -sd_1, \alpha_2$	0.59 ± 0.14
11	$-mRR, r_{nRR}, std_f, std_{mRR}, VLF, LF, nLF, -nHF, -sd_1, \alpha_1, \alpha_2$	0.57 ± 0.14
12	$-mRR, r_{nRR}, std_f, std_{mRR}, VLF, LF, LFHF, -nHF, -sd_1, sd_2, \alpha_1, \alpha_2,$	0.56 ± 0.13
13	$-mRR, r_{nRR}, std_f, std_{mRR}, VLF, LF, HF, LFHF, -nHF, -sd_1, sd_2, \alpha_1, \alpha_2,$	0.54 ± 0.13
14	All parameters	0.53 ± 0.14

Note: results are presented with mean $\pm$ standard deviation.

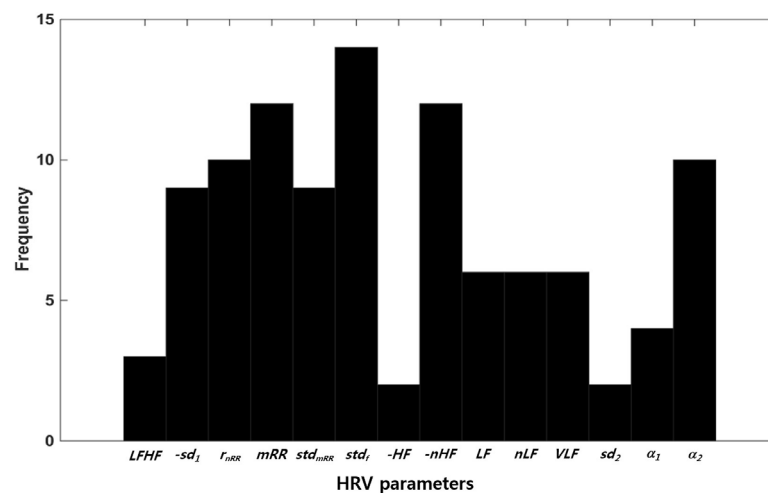


Figure 5. Frequency of selected parameters used in each combination.

3.2. Performance evaluation

The estimated REM sleep using the proposed algorithm was compared with the scoring result from the reference PSG. As shown in table 4, the algorithm detected REM sleep with average Cohen's kappa coefficients of 0.63 and 0.61, accuracies of 87.08% and 87.03%, sensitivities of 78.98% and 80.01%, and specificities of 89.27% and 88.67% for the training and validation sets, respectively. Figure 7 shows examples of the scoring results obtained by the automatic algorithm and the PSG of healthy and OSA (AHI: 19.4) subjects. An example of a normal subject is presented in figure 7(a), which provided a Cohen's kappa coefficient of 0.62,

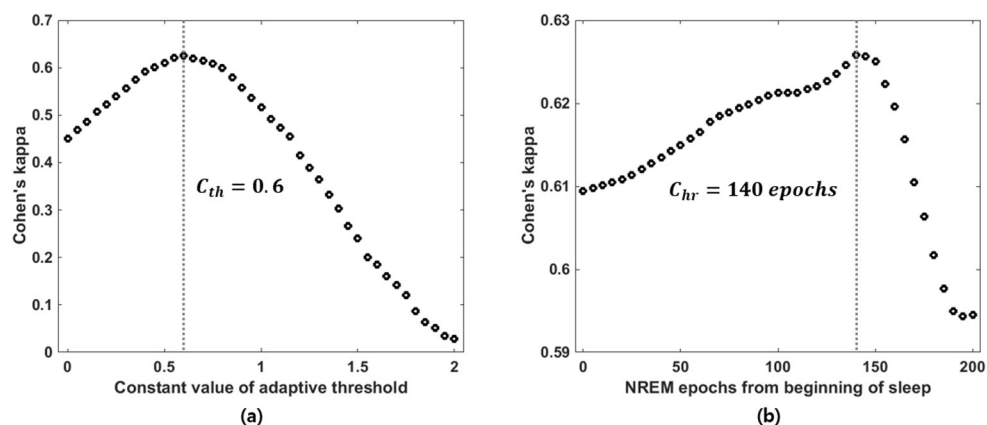


Figure 6. Determination of (a) the constant value (C_{th}) applied to the adaptive threshold and (b) the time span (C_{hr}) for NREM sleep after the beginning of sleep.

Table 4. Statistical results for detecting REM sleep based on an epoch-by-epoch comparison.

Set	Group	N	k	Accuracy (%)	Sensitivity (%)	Specificity (%)
Training	Healthy	10	0.65 ± 0.11	87.90 ± 4.00	83.85 ± 10.24	89.14 ± 4.78
	OSA	16	0.61 ± 0.13	86.57 ± 5.18	75.93 ± 11.03	89.35 ± 5.49
	Total	26	0.63 ± 0.13	87.08 ± 4.73	78.98 ± 11.23	89.27 ± 5.13
Validation	Healthy	11	0.63 ± 0.12	87.11 ± 4.96	83.29 ± 8.88	88.37 ± 5.40
	OSA	14	0.59 ± 0.15	86.97 ± 4.29	77.43 ± 11.12	88.91 ± 4.77
	Total	25	0.61 ± 0.13	87.03 ± 4.49	80.01 ± 10.43	88.67 ± 4.95

OSA: obstructive sleep apnea, N : number of subjects, k : Cohen's kappa.

Note: results are presented with mean $\pm$ standard deviation.

All statistical results were not significantly different between healthy and OSA groups in both training and validation sets ($p > 0.05$).

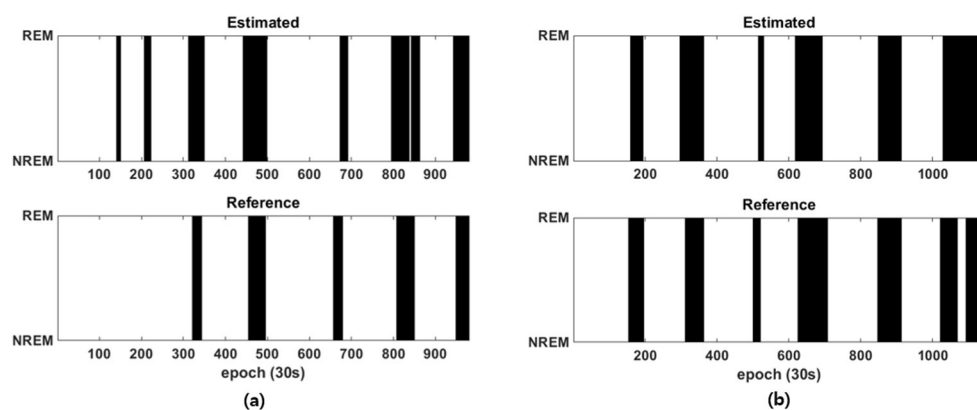


Figure 7. Examples of epoch-by-epoch scoring results by the proposed algorithm and reference PSG from (a) a normal subject and (b) an OSA (AHI: 19.4) subject.

Table 5. Comparison of the REM sleep-related information from the PSG and algorithm.

Set	Duration of REM sleep (min)			Percentage of REM sleep (%)		
	Est.	Ref.	Corr.	Est.	Ref.	Corr.
Training	126.42 ± 30.37	100.75 ± 26.59	0.56 ^c	25.42 ± 5.18	20.24 ± 4.68	0.39 ^a
Validation	124.98 ± 28.20	93.98 ± 29.46	0.58 ^c	25.09 ± 4.78	18.85 ± 5.25	0.47 ^b

Est.: Estimated, Ref.: Reference, Corr.: Correlation coefficient.

^a $p < 0.05$.

^b $p < 0.02$.

^c $p < 0.003$.

an accuracy of 87.54%, a sensitivity of 87.26%, and a specificity of 89.59%. The corresponding values from the OSA subject in figure 7(b) were 0.77, 90.13%, 86.93%, and 91.55%, respectively.

3.3. Statistical analysis

A comparison of the performance indices between the training and validation sets was conducted to evaluate the statistical differences. As a result, there were no statistical differences in the Cohen's kappa coefficient, accuracy, sensitivity, and specificity between the training and validation sets (Mann–Whitney U test: $p > 0.05$ for all performance values). Moreover, a statistical analysis of the corresponding values between healthy subjects and the patients with OSA in the training and validation sets was performed. Statistical differences in the performance indices were not detected between the healthy and OSA groups in both the training and validation sets (Mann–Whitney U test: $p > 0.05$ for all performance values). From the results of automatic scoring, REM sleep-related information was obtained and compared with that from PSG, as summarized in table 5. The REM sleep-related information extracted from the proposed algorithm showed a significant correlation with the corresponding parameter from the PSG in both the training and validation sets.

4. Discussion

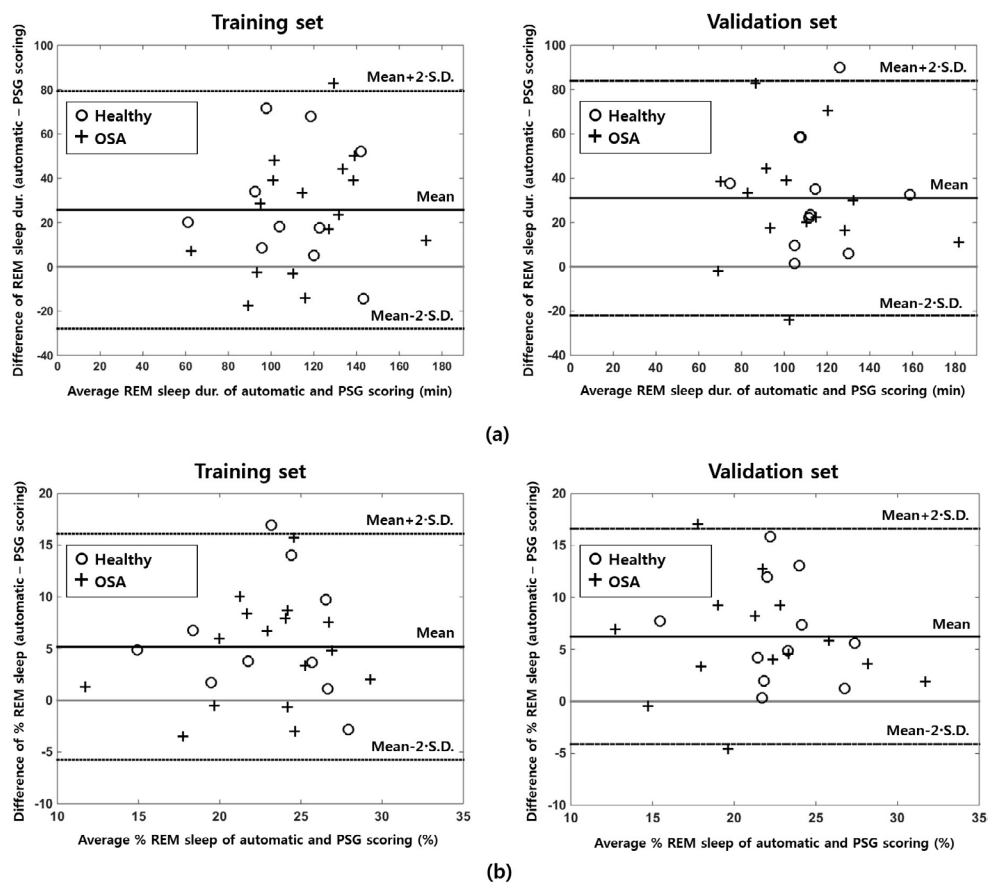
4.1. Agreement between the proposed algorithm and reference PSG

In this study, we suggested a method for detecting REM sleep using R–R intervals. The algorithm was developed using the data from 26 healthy and OSA subjects and was evaluated with those from a separate set of 25 healthy and OSA subjects. Since it is unknown whether or not the user is a patient with sleep-related disorders in practical use, the algorithm was established and validated with the data from healthy subjects and the patients with OSA, which is one of the major sleep-related disorders. As a result, the Cohen's kappa statistics were greater than 0.6, which revealed substantial agreement (0.61–0.80) in both the training and validation sets according to the epoch-by-epoch analysis. In addition, the performance analysis values—the accuracy, sensitivity, specificity, and Cohen's kappa—were not significantly different between the healthy and OSA groups in the training and validation sets. The epoch-by-epoch comparison of the scoring results between the proposed algorithm and the PSG is presented as a confusion matrix in table 6. Approximately 33% of the false positives (FPs) were derived from awakening epochs that were incorrectly detected as REM sleep by the proposed method. Because the extracted feature from the HRV parameters was smoothed before REM sleep detection, it was difficult to distinguish awakening epochs during an REM sleep period from actual REM sleep. In addition, the long-term awakening epochs during which

Table 6. A 2×2 confusion matrix of automatic REM sleep detection compared with PSG.

Set	Estimated	Reference (PSG)	
		REM	NREM
Training	REM	4125	2218 (730)
	NREM	1114	18311
Validation	REM	3769	2319 (766)
	NREM	930	17862

Note: numbers in parentheses indicate the number of awakening epochs that were incorrectly detected as REM sleep by the algorithm.

**Figure 8.** Bland–Altman plot of (a) the REM sleep duration and (b) the percentage of REM sleep in the training and validation sets. S.D.: standard deviation, dur.: duration, min: minute.

sympathetic activities may increase could be estimated as REM sleep. Moreover, the NREM periods preceding and following each REM period were slightly overestimated as REM sleep. This result influenced the agreement of the REM sleep-related information between the proposed algorithm and the PSG. As shown in figure 8, the Bland–Altman plot analysis revealed that each parameter was slightly overestimated. The mean differences in the duration of REM

sleep were 25.7 and 31.0 min, and percentages of REM sleep were 5.2% and 6.2% in training and validation sets, respectively. Nevertheless, the results revealed substantial agreement (0.61–0.80) in the Cohen's kappa statistics, and there were significant correlations between the estimated and reference REM sleep information ($p < 0.05$ for all parameters in both the training and validation sets). It is known that manual PSG scoring is quite time consuming and a laborious process. In addition, several studies have revealed the inter- and intra-rater variabilities in manual PSG scoring. As for the scoring of REM sleep, the inter-rater agreement has been reported to be in the range of 75–98% (Silber *et al* 2007, Danker-hopfe *et al* 2009), and the intra-rater agreement examined for three scores with a rescoring time interval of 6.5 months was in the range of 72–88% (Whitney *et al* 1998). Compared with these results, our result obtained by automatic scoring only using R–R intervals is fairly competitive. In addition, the overestimation could be somewhat improved by combining with previously developed awakening detection algorithms. In addition, the proposed method with the awakening detection algorithm could provide more accurate information associated with REM sleep.

4.2. Evaluation of the REM sleep detection algorithm

REM sleep was automatically detected on the basis of the following procedures: (1) calculating the HRV parameters using R–R intervals, (2) extracting an REM feature based on a PCA, (3) determining and applying an adaptive threshold, and (4) applying a heuristic rule. It is known that the sleep stages are associated with autonomic nervous activities (Verrier and Harper 2011, Bartsch *et al* 2012). Thus, related physiological signals such as heart and respiratory rhythms have been shown to have different patterns according to the sleep states (Galletly and Larsen 1998, Heinzer and Series 2011, Lanfranchi and Somers 2011). Therefore, it is appropriate to estimate the sleep stages on the basis of these characteristics. REM sleep is highlighted by the fluctuation between sympathetic and parasympathetic activities. The conventional time-domain, frequency-domain, and nonlinear HRV parameters have been reported to have significant differences between NREM and REM sleep (Bunde *et al* 2000, Penzel *et al* 2003, Lanfranchi and Somers 2011). To estimate REM sleep, we extracted not only the well-known HRV parameters but also newly developed parameters. std_{mRR} is the standard deviation for the mean of the R–R intervals (mRR) within 5 symmetrically adjacent epochs. Thus, an increase representing the sudden and irregular changes in the heart rate during REM sleep was expected. r_{nRR} is the correlation coefficient between the normalized R–R intervals (nRRI) and the smoothed nRRI. As shown in figure 1, the fluctuation in the R–R intervals is predominant during REM sleep; therefore, r_{nRR} was higher compared with that in NREM sleep. In addition, the respiratory influence is also parameterized from the R–R interval. On the basis of RSA phenomenon (Angelone and Coulter 1964), the respiratory signal was obtained from the R–R intervals; then, the respiratory frequency was calculated in a 30 s window. std_f is the standard deviation of the mean of the respiratory frequencies within 5 adjacent epochs, representing the irregular patterns of the respiratory rhythm.

One of difficulties for sleep estimation using the HRV parameters is that various factors can influence cardiorespiratory activities (Long *et al* 2015). That is, even though major fluctuations in the parameters are associated with the sleep cycle, the levels of these patterns can vary, and even the fluctuations related to the sleep cycle may not be always guaranteed. Therefore, several strategies were considered for extracting an REM feature and determining the threshold to establish a suitable model for REM sleep estimation. First, an REM feature was determined with a combination of HRV parameters representing different characteristics of the cardiorespiratory system using a PCA. A PCA provides principal components that are linearly uncorrelated, and the first principal component (PC_1) has the largest variation. Hence, PC_1 based on the HRV parameters can be represented a major fluctuation, which corresponds with

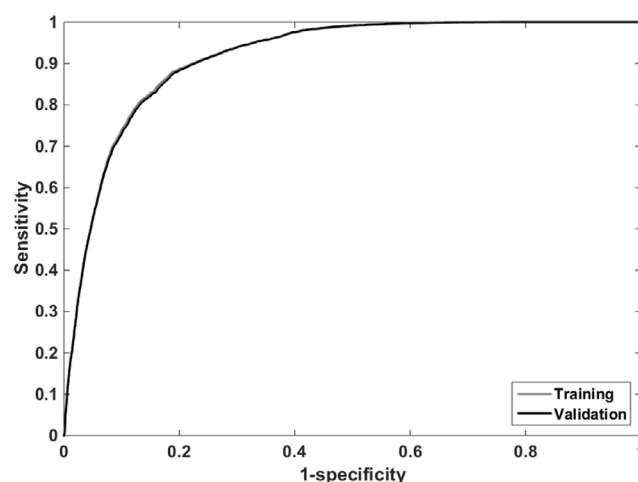


Figure 9. REM detection accuracy analysed with the receiving–operating characteristic curve (ROC) for the training and validation data.

the sleep cycle. In addition, PCA was also appropriate method to extract an REM feature when at least several parameters reflect the sleep-cycle-related fluctuation. However, it is not known which combination of parameters provides the best performance for REM sleep detection. Therefore, an optimal set of HRV parameters was chosen to maximize Cohen's kappa using the training data of 26 subjects. The best result for REM sleep estimation was confirmed when 7 HRV parameters were used, as summarized in table 3. In addition, the 7 selected parameters were also frequently used in other combinations, as illustrated in figure 5. Therefore, the combination of the 7 parameters was suitable for extracting the feature representing the REM characteristics. Second, the threshold for REM sleep determination was not constant but a trend of the extracted feature. Therefore, it was adaptively changed to reflect the patterns of the REM sleep cycle. Finally, a heuristic rule was applied to modify the detected REM sleep. During normal sleep, REM sleep rarely appears 70–100 min after the beginning of sleep (Carskadon and Dement 2011). On the basis of this fact, REM sleep estimated within several minutes after the beginning of sleep was regarded as NREM sleep. All specific values used in the threshold determination and heuristic rule were also optimized to maximize Cohen's kappa using the training data, as shown in figure 6. As a result, the values of Cohen's kappa were 0.63 and 0.61 for the training and validation sets, respectively, which reveal greater than substantial agreement. In addition, there were no performance differences between the normal and OSA groups in both the training and validation sets, as summarized in table 4. Finally, the algorithm using 7 HRV parameters were analysed with a receiving–operating characteristic curve (ROC). As shown in figure 9, the areas under the curve (AUCs) were 92% and 91% for the training and validation data, respectively. The proposed processes provided a specific method for feature extraction and event determination. Hence, they can be applied to various research fields where needed such as PSG data analysis.

4.3. Comparison with previous studies

REM sleep estimation based on ANS-related signals has been presented in various studies, as summarized in table 7. One study reported that the amplitudes of the peripheral arterial tone (PAT) increase during REM sleep because of peripheral vasoconstriction (Lavie *et al* 2000).

Table 7. Comparison with related works.

Author (year)	Signals	<i>N</i>	<i>k</i>	Acc. (%)	Group
Herscovici (2006)	Peripheral arterial tone	30	0.58	89	Healthy, OSA
Hedner (2011)	Peripheral arterial tone	227	0.55	88	Healthy, OSA
Willemen (2014) ^a	ECG, movement, thoracic/abdominal volume change	28	0.56	86	Healthy
Xiao (2013)	ECG	45	0.50	60	Healthy
Tanida (2013)	ECG	23	—	80	Healthy
Kortelainen (2010)	BCG, movement	18	—	80	Healthy
Watanabe (2004)	BCG, movement	12	—	38	Healthy
Chung (2009)	Nasal airflow	22	0.63	88	Healthy, OSA
Chung (2009)	Thoracic volume change	22	0.62	88	Healthy, OSA
Our work	ECG	25	0.61	87	Healthy, OSA

N: number of validated data sets, *k*: Cohen's kappa, Acc.: accuracy, ECG: electrocardiogram, BCG: ballistocardiogram.

^a Min-by-min analysis.

On the basis of this finding, Herscovici *et al* developed an automatic algorithm to estimate REM sleep (Herscovici *et al* 2006). From the signal recorded using a Watch-PAT100 device, the PAT- and pulse-rate-related parameters were extracted. On the basis of a genetic algorithm, the overall Cohen's kappa and accuracy for detecting REM sleep were 0.58 and 89% for data from 30 healthy and OSA subjects. The same algorithm was validated with 227 data sets from 3 different sleep centres. The results provided a Cohen's kappa coefficient of 0.55 and an accuracy of 88% (Hedner *et al* 2011). Willemen *et al* established a method to classify sleep into four stages: wakefulness, N1–N2, N3, and REM sleep on the basis of a support vector machine classifier (Willemen *et al* 2014). The heart and breathing rates were extracted from an ECG and respiratory inductance plethysmography, respectively, and body movement was measured using a 2D mattress-type sensor. On the basis of this information, the parameters were calculated and used for automatic sleep staging. According to a min-by-min analysis, the performance for REM sleep classification was a Cohen's kappa coefficient of 0.56 and an accuracy of 86% for the validation data of 28 healthy subjects. Xiao *et al* developed an automated sleep monitoring algorithm using the HRV parameters extracted from an ECG on the basis of a random forest method; the performance results were a Cohen's kappa coefficient of 0.50 and an accuracy of 60% for REM sleep estimation (Xiao *et al* 2013). Another study also assessed automatic sleep staging only using an ECG (Tanida *et al* 2013). According to rule-based decision, sleep was discriminated in four stages. The results based on 23 validation data sets revealed an accuracy of 80% for REM sleep detection. Several researchers developed a sleep staging method using the data measured from non-contact sensors. A BCG is a measure of the recoil forces of the body derived from cardiac ejection (Starr *et al* 1939). Because a heart-induced vibration can be transmitted through the medium, the heart rate can be measured using pressure sensors or load cells without direct skin contact (Choi *et al* 2009, Inan *et al* 2009, Inan *et al* 2015). Using a BCG and body movement, the performance of REM sleep detection was an accuracy of 38% for 12 validation data

sets (Watanabe and Watanabe 2004) and an accuracy of 80% for 18 validation data sets (Kortelainen *et al* 2010) in separate studies. Chung *et al* developed models to discriminate REM and NREM sleep using respiratory dynamics, which is a representative characteristic of REM sleep (Chung *et al* 2009). The methods were validated with two different respiratory signals—the nasal airflow with a thermocouple and the thoracic volume change with a piezoelectric-type belt—from 22 healthy and OSA subjects. The performance results were Cohen's kappa coefficients of 0.63 and 0.62 and accuracies of 88% and 88% for each use of a respiratory signal. In this study, we developed and validated the algorithm to detect REM sleep only using R–R intervals according to an epoch-by-epoch analysis. The algorithm was trained and validated with healthy subjects and patients with OSA. In practical use, it is not known whether or not the user is a patient with sleep disorders. Therefore, we established an algorithm that can be applied to both groups. Finally, the performance of our method for REM sleep estimation was quite competitive compared with that in previous studies. In addition, the R–R intervals can be widely acquired with wearable and mobile devices that measure a formal ECG or alternative signals such as the PPG, BCG, capacitive ECG, and so on. Therefore, the proposed algorithm can be utilized with various devices; thus, it is appropriate for home-based long-term sleep monitoring.

4.4. Limitations

This study has several limitations that should be considered in future study. First, the automatic algorithm to determine REM sleep was developed and evaluated using the data from 51 healthy subjects and patients with OSA. Even though more than 23 000 epochs were used to establish and validate the algorithm, respectively, the cohort size was still not sufficient to cover various ranges in demographics and sleep variables. However, it is necessary to consider the variables in a large variation range for practical applications because factors such as age, BMI, and AHI affect the sleep architecture (Redline *et al* 2004), which may influence the performance of the proposed algorithm. In addition, the groups of subjects did not contain other sleep-related disorders such as insomnia and periodic limb movement syndrome, which are known to be linked with sympathetic nervous activities (Bonnet and Arand 1998, Guggisberg *et al* 2007). They may influence the detection of REM sleep using the proposed method, which is based on the autonomic nervous activity measured by HRV. Therefore, it is necessary to validate the proposed method with a large amount of data from patients with sleep-related disorders. Second, our study considered 70 min after the beginning of sleep as the NREM period on the basis of a heuristic rule. Even though the rule was based on the statistics of normal human sleep (Carskadon and Dement 2011) and a specific time span (70 min) was determined by an optimization process, it has been reported that a short REM latency—the time interval between the onset of sleep and the beginning of the first REM sleep—is one of the characteristics of patients with narcolepsy and endogenous depression (Guilleminault and Cao 2011, Peterson and Benca 2011). Therefore, it is necessary to modify the restriction of the detection of REM sleep within 70 min after the beginning of recording. Third, each PSG data was scored by one technician even though the result was reviewed by clinicians. However, a more reliable comparison between the results from the automatic algorithm and those from the PSG could be achieved if the inter- and intra-rater variabilities are considered in the PSG scoring. Finally, one merit of our method is the detection of REM sleep using only R–R intervals. It can be combined with commercial devices that record heart-rate-related signals. In a future study, the algorithm should be validated with data recorded at home and in ambulatory environments to confirm its usability and significance.

5. Conclusions

In this study, an automatic algorithm for detecting REM sleep was established on the basis of the autonomic dynamics extracted from R–R intervals. Even though there are several limitations, the proposed method exhibited persuasive results for the detection of REM sleep compared with those in comparable studies. The use of only R–R intervals allows the proposed algorithm to be easily compatible with mobile and wearable devices that provide heart-rhythm-associated data. Therefore, the method is appropriate for long-term sleep monitoring in the home and ambulatory circumstances when standard PSG is not accessible. In the future, by combining the proposed algorithm with algorithms that detect other stages such as wakefulness and SWS, detailed sleep information can be available. The data obtained during long-term monitoring could provide useful information to healthy people and patients with sleep disorders for health management. In addition, the data could be also used to monitor the improvements in patient's sleep after diagnosis and treatments for the clinical purposes.

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